

Solutions to Quantum Mechanics: Concepts and
Applications
by Nouredine Zettili

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*To Dr. Miller Mendoza Jiménez from ETH Zürich
Thank you for inspiring me to do theoretical physics.*

Preface

This is an ongoing project of mine to solve a significant percentage of all the problems in the third edition of Zettili's *Quantum Mechanics* [Zet22]. I frequently refer to [Axl24], due to the necessity of linear algebra in quantum theory. As always, should there be any errors, please email me:

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The only way to understand physics is by *doing* physics. However, working your way through problems is useless without the ability to verify your solutions. There is good reason why books on the subject (usually) do not make the answers public, for immediately looking at the answers the moment you get stuck on a problem is extremely detrimental to one's comprehension of a topic. That said, it is also detrimental to an individual's understanding if they do not have the ability to verify their workings. Therefore, it is imperative that you, the reader, use this manual *wisely*, and do not give up on a difficult problem. Of course, if you really can't solve it, do consult the solutions, but always remember that the lesson of an exercise is not the exercise itself, but rather the way it forces you to organise your thoughts before solving it.

I hope that this manuscript becomes useful to the theoretical physics community (undergraduate students) as I continue to update it with more and more solutions.

Kieron Eisma

Chapter 1

Origins of Quantum Physics

Exercise 1.37.

Consider the function

$$\psi(x) = \begin{cases} 1 - |x|, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

To find the Fourier Transform $\phi(p)$ of this function, consult Exercise 2 on page 161 of [SS03].

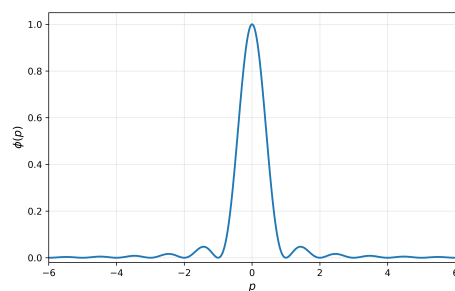


Figure 1.1: *Plot of the Fourier transform $\phi(p)$ of ψ . Plot generated by ChatGPT*

Chapter 2

Mathematical Tools of Quantum Mechanics

Exercise 2.1(a)

For the first two norms, we have:

$$\begin{aligned}\langle\psi|\psi\rangle &= (-i\langle\phi_1| - 3i\langle\phi_2| - \langle\phi_3|)(i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle) \\ &= (-i)(i)\langle\phi_1|\phi_1\rangle + (-i)(3i)\langle\phi_1|\phi_2\rangle + (-i)(-1)\langle\phi_1|\phi_3\rangle \\ &\quad + (-3i)(i)\langle\phi_2|\phi_1\rangle + (-3i)(3i)\langle\phi_2|\phi_2\rangle + (-3i)(-1)\langle\phi_2|\phi_3\rangle \\ &\quad + (-1)(i)\langle\phi_3|\phi_1\rangle + (-1)(3i)\langle\phi_3|\phi_2\rangle + (-1)(-1)\langle\phi_3|\phi_3\rangle \\ &= 1 + 9 + 1 \\ &= 11,\end{aligned}$$

and

$$\begin{aligned}\langle\chi|\chi\rangle &= (\langle\phi_1| + i\langle\phi_2| - 5i\langle\phi_3|)(|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle) \\ &= \langle\phi_1|\phi_1\rangle + \langle\phi_2|\phi_2\rangle + 25\langle\phi_3|\phi_3\rangle \\ &= 1 + 1 + 25 \\ &= 27.\end{aligned}$$

For a much faster method to calculate these norms, see 6.30 (b) in [Axl24]

Now, for $\langle\chi|\psi\rangle$ and $\langle\psi|\chi\rangle$:

$$\begin{aligned}
\langle \chi | \psi \rangle &= (\langle \phi_1 | + i\langle \phi_2 | - 5i\langle \phi_3 |)(i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle) \\
&= i\langle \phi_1 | \phi_1 \rangle - 3\langle \phi_2 | \phi_2 \rangle + 5i\langle \phi_3 | \phi_3 \rangle \\
&= i - 3 + 5i \\
&= -3 + 6i,
\end{aligned}$$

and

$$\begin{aligned}
\langle \psi | \chi \rangle &= (-i\langle \phi_1 | - 3i\langle \phi_2 | - \langle \phi_3 |)(|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle) \\
&= -i\langle \phi_1 | \phi_1 \rangle - 3\langle \phi_2 | \phi_2 \rangle - 5i\langle \phi_3 | \phi_3 \rangle \\
&= -i - 3 - 5i \\
&= -3 - 6i.
\end{aligned}$$

Note that the line of working for $\langle \psi | \chi \rangle$ is redundant because we could have taken the complex conjugate of $\langle \chi | \psi \rangle$.

As expected, we have that $\langle \chi | \psi \rangle = \langle \psi | \chi \rangle^*$.

For $\langle \psi + \chi | \psi + \chi \rangle$:

$$\begin{aligned}
\langle \psi + \chi | \psi + \chi \rangle &= \langle \psi | \psi \rangle + \langle \psi | \chi \rangle + \langle \chi | \psi \rangle + \langle \chi | \chi \rangle \\
&= 11 + (-3 - 6i) + (-3 + 6i) + 27 \\
&= 32.
\end{aligned}$$

Exercise 2.1(b)

To compute the operators $|\psi\rangle\langle\chi|$ and $|\chi\rangle\langle\psi|$:

$$\begin{aligned}
|\psi\rangle\langle\chi| &= (i|\phi_1\rangle + 3i|\phi_2\rangle - |\phi_3\rangle)(\langle\phi_1| + i\langle\phi_2| - 5i\langle\phi_3|) \\
&= i|\phi_1\rangle\langle\phi_1| + i^2|\phi_1\rangle\langle\phi_2| - 5i^2|\phi_1\rangle\langle\phi_3| \\
&\quad + 3i|\phi_2\rangle\langle\phi_1| + 3i^2|\phi_2\rangle\langle\phi_2| - 15i^2|\phi_2\rangle\langle\phi_3| \\
&\quad - |\phi_3\rangle\langle\phi_1| - i|\phi_3\rangle\langle\phi_2| + 5i|\phi_3\rangle\langle\phi_3| \\
&= i|\phi_1\rangle\langle\phi_1| - |\phi_1\rangle\langle\phi_2| + 5|\phi_1\rangle\langle\phi_3| \\
&\quad + 3i|\phi_2\rangle\langle\phi_1| - 3|\phi_2\rangle\langle\phi_2| + 15|\phi_2\rangle\langle\phi_3| \\
&\quad - |\phi_3\rangle\langle\phi_1| - i|\phi_3\rangle\langle\phi_2| + 5i|\phi_3\rangle\langle\phi_3|,
\end{aligned}$$

and

$$\begin{aligned}
|\chi\rangle\langle\psi| &= (|\phi_1\rangle - i|\phi_2\rangle + 5i|\phi_3\rangle)(-i\langle\phi_1| - 3i\langle\phi_2| - \langle\phi_3|) \\
&= -i|\phi_1\rangle\langle\phi_1| - 3i|\phi_1\rangle\langle\phi_2| - |\phi_1\rangle\langle\phi_3| \\
&\quad - i(-i)|\phi_2\rangle\langle\phi_1| - 3i(-i)|\phi_2\rangle\langle\phi_2| + i|\phi_2\rangle\langle\phi_3| \\
&\quad - 5i^2|\phi_3\rangle\langle\phi_1| - 15i^2|\phi_3\rangle\langle\phi_2| - 5i|\phi_3\rangle\langle\phi_3| \\
&= -i|\phi_1\rangle\langle\phi_1| - 3i|\phi_1\rangle\langle\phi_2| - |\phi_1\rangle\langle\phi_3| \\
&\quad - |\phi_2\rangle\langle\phi_1| - 3|\phi_2\rangle\langle\phi_2| + i|\phi_2\rangle\langle\phi_3| \\
&\quad + 5|\phi_3\rangle\langle\phi_1| + 15|\phi_3\rangle\langle\phi_2| - 5i|\phi_3\rangle\langle\phi_3|.
\end{aligned}$$

The traces are easy to compute, because $\text{tr}(\hat{A}\hat{B}) = \text{tr}(\hat{B}\hat{A})$ (for a proof, see 8.56 in [Axl24]):

$$\text{tr}(|\psi\rangle\langle\chi|) = \text{tr}(\langle\chi|\psi\rangle) = -3 + 6i$$

$$\text{tr}(|\chi\rangle\langle\psi|) = \text{tr}(\langle\psi|\chi\rangle) = -3 - 6i$$

Notice how the traces are complex conjugates of one another.

Exercise 2.1 (c)

The computation of the requested Hermitian conjugates of the quantum states are trivial.

Exercise 2.2

Consider the state vector $|\psi_1\rangle = |\phi_1\rangle + 4i|\phi_2\rangle + 5|\phi_3\rangle$. Taking complex conjugates and interchanging kets with bras, we obtain

$$\langle\psi_1| = \langle\phi_1| - 4i\langle\phi_2| + 5\langle\phi_3|,$$

Since $|\psi_2\rangle = b|\phi_1\rangle + 4|\phi_2\rangle - 3i|\phi_3\rangle$, we compute the inner product

$$\begin{aligned}
\langle\psi_1|\psi_2\rangle &= (\langle\phi_1| - 4i\langle\phi_2| + 5\langle\phi_3|)(b|\phi_1\rangle + 4|\phi_2\rangle - 3i|\phi_3\rangle) \\
&= b\langle\phi_1|\phi_1\rangle + 4\langle\phi_1|\phi_2\rangle - 3i\langle\phi_1|\phi_3\rangle \\
&\quad - 4ib\langle\phi_2|\phi_1\rangle - 16i\langle\phi_2|\phi_2\rangle + 12\langle\phi_2|\phi_3\rangle \\
&\quad + 5b\langle\phi_3|\phi_1\rangle + 20\langle\phi_3|\phi_2\rangle - 15i\langle\phi_3|\phi_3\rangle \\
&= b - 16i - 15i \\
&= b - 31i.
\end{aligned}$$

For the quantum states to be orthogonal, require that $\langle \psi_1 | \psi_2 \rangle := 0$, which forces $b = 31i$ by the equation above.

Again, for a much quicker calculation, see 6.30(c) in [Axl24].

Exercise 2.3

All the necessary products for you to verify (a) the Cauchy-Schwarz Inequality and (b) the Triangle Inequality are contained in Exercise 2.1(a).

These inequalities must always be true, by 6.14 and 6.17 in [Axl24], so the result is no surprise.

Exercise 2.4

Consider the quantum states

$$\begin{aligned} |\psi\rangle &= \alpha|\phi_1\rangle + 5|\phi_2\rangle \\ |\chi\rangle &= 3\alpha|\phi_1\rangle - 4|\phi_2\rangle. \end{aligned}$$

Taking the Hermitian conjugate of the state $|\psi\rangle$ yields the bra

$$\langle\psi| = \bar{\alpha}\langle\phi_1| + 5\langle\phi_2|,$$

so that

$$\begin{aligned} \langle\psi|\chi\rangle &= (\bar{\alpha}\langle\phi_1| + 5\langle\phi_2|)(3\alpha|\phi_1\rangle - 4|\phi_2\rangle) \\ &= 3\alpha\bar{\alpha}\langle\phi_1|\phi_1\rangle - 4\bar{\alpha}\langle\phi_1|\phi_2\rangle \\ &\quad + 15\alpha\langle\phi_2|\phi_1\rangle - 20\langle\phi_2|\phi_2\rangle \\ &= 3|\alpha|^2 - 20. \end{aligned}$$

In order for the states $|\psi\rangle$ and $|\chi\rangle$ to be orthogonal, mandate

$$\langle\psi|\chi\rangle = 0,$$

which forces

$$|\alpha|^2 = \frac{20}{3}.$$

Now for $\alpha \in \mathbb{R}$, we can either have $\alpha = \sqrt{\frac{20}{3}}$ or $\alpha = -\sqrt{\frac{20}{3}}$.

More generally, observe that, if $B(x, r)$ denotes the open ball of radius r around a point x in a metric space (X, d) , we see that for all

$$\alpha \in \partial B\left(\mathbf{0}_{\mathbb{C}}, \sqrt{20/3}\right),$$

the states $|\psi\rangle$ and $|\chi\rangle$ will be orthonormal.

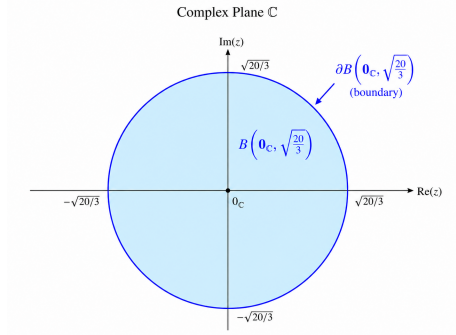


Figure 2.1: The complex ball centered about the origin (image generated by ChatGPT)

Exercise 2.5

Consider the quantum states $|\psi\rangle = |\phi_1\rangle + |\phi_2\rangle$ and $|\chi\rangle = |\phi_1\rangle - |\phi_2\rangle$, where $|\phi_1\rangle$ and $|\phi_2\rangle$ are *not* orthonormal. We claim that we have the following identities:

$$\begin{aligned}\langle\psi|\psi\rangle + \langle\chi|\chi\rangle &= 2\langle\phi_1|\phi_1\rangle + 2\langle\phi_2|\phi_2\rangle, \\ \langle\psi|\psi\rangle - \langle\chi|\chi\rangle &= 2\langle\phi_1|\phi_2\rangle + 2\langle\phi_2|\phi_1\rangle.\end{aligned}$$

Proof:

Taking the Hermitian conjugate of the state vector $|\psi\rangle = |\phi_1\rangle + |\phi_2\rangle$ yields the bra $\langle\psi| = \langle\phi_1| + \langle\phi_2|$. Consequently, the norm of $|\psi\rangle$ is given by

$$\begin{aligned}\langle\psi|\psi\rangle &= (\langle\phi_1| + \langle\phi_2|)(|\phi_1\rangle + |\phi_2\rangle) \\ &= \langle\phi_1|\phi_1\rangle + \langle\phi_1|\phi_2\rangle + \langle\phi_2|\phi_1\rangle + \langle\phi_2|\phi_2\rangle.\end{aligned}$$

Similarly, we have $\langle\chi| = \langle\phi_1| - \langle\phi_2|$,

which yields

$$\begin{aligned}\langle\chi|\chi\rangle &= (\langle\phi_1| - \langle\phi_2|)(|\phi_1\rangle - |\phi_2\rangle) \\ &= \langle\phi_1|\phi_1\rangle - \langle\phi_1|\phi_2\rangle - \langle\phi_2|\phi_1\rangle + \langle\phi_2|\phi_2\rangle.\end{aligned}$$

It immediately follows that

$$\begin{aligned}
\langle \psi | \psi \rangle + \langle \chi | \chi \rangle &= (\langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle + \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&\quad + (\langle \phi_1 | \phi_1 \rangle - \langle \phi_1 | \phi_2 \rangle - \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&= 2\langle \phi_1 | \phi_1 \rangle + 2\langle \phi_2 | \phi_2 \rangle,
\end{aligned}$$

and

$$\begin{aligned}
\langle \psi | \psi \rangle - \langle \chi | \chi \rangle &= (\langle \phi_1 | \phi_1 \rangle + \langle \phi_1 | \phi_2 \rangle + \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&\quad - (\langle \phi_1 | \phi_1 \rangle - \langle \phi_1 | \phi_2 \rangle - \langle \phi_2 | \phi_1 \rangle + \langle \phi_2 | \phi_2 \rangle) \\
&= 2\langle \phi_1 | \phi_2 \rangle + 2\langle \phi_2 | \phi_1 \rangle,
\end{aligned}$$

which is the desired result. ■

Exercise 2.6(a)

Consider the quantum state

$$|\psi\rangle = \frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle$$

where the kets $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ are orthonormal.

By orthonormality, we have that

$$\begin{aligned}
\langle \psi | \psi \rangle &= \left(\frac{1}{\sqrt{15}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{5}} \right)^2 \\
&= \frac{1}{15} + \frac{1}{3} + \frac{1}{5} \\
&= \frac{9}{15} \\
&= \frac{3}{5}.
\end{aligned}$$

It follows the norm of $|\psi\rangle$ is given by

$$\|\psi\| = \sqrt{\frac{3}{5}}.$$

Exercise 2.6(b)

Given further that the $|\phi_n\rangle$ are the eigenstates of an operator $\hat{B}$, with $\hat{B}|\phi_n\rangle = (3n^2 - 1)|\phi_n\rangle$, for $n \in \{1, 2, 3\}$.

By linearity, one can verify that

$$\begin{aligned}
 \hat{B}|\psi\rangle &= \hat{B}\left(\frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{1}{\sqrt{15}}\hat{B}|\phi_1\rangle + \frac{1}{\sqrt{3}}\hat{B}|\phi_2\rangle + \frac{1}{\sqrt{5}}\hat{B}|\phi_3\rangle \\
 &= \frac{2}{\sqrt{15}}|\phi_1\rangle + \frac{11}{\sqrt{3}}|\phi_2\rangle + \frac{26}{\sqrt{5}}|\phi_3\rangle.
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \langle\psi|\hat{B}|\psi\rangle &= \left(\frac{1}{\sqrt{15}}\langle\phi_1| + \frac{1}{\sqrt{3}}\langle\phi_2| + \frac{1}{\sqrt{5}}\langle\phi_3|\right)\left(\frac{2}{\sqrt{15}}|\phi_1\rangle + \frac{11}{\sqrt{3}}|\phi_2\rangle + \frac{26}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{2}{15} + \frac{11}{3} + \frac{26}{5} \\
 &= \frac{2 + 55 + 78}{15} \\
 &= \frac{135}{15} \\
 &= 9.
 \end{aligned}$$

The expectation value of $\hat{B}$ with respect to the state $|\psi\rangle$ is then

$$\begin{aligned}
 \langle\hat{B}\rangle &= \frac{\langle\psi|\hat{B}|\psi\rangle}{\langle\psi|\psi\rangle} \\
 &= \frac{9}{3/5} \\
 &= 15.
 \end{aligned}$$

where the second equality is a consequence of our work in part (a).

Exercise 2.6(c)

$$\begin{aligned}
 \hat{B}^2|\psi\rangle &= \hat{B}^2\left(\frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle\right) \\
 &= \frac{1}{\sqrt{15}}\hat{B}^2|\phi_1\rangle + \frac{1}{\sqrt{3}}\hat{B}^2|\phi_2\rangle + \frac{1}{\sqrt{5}}\hat{B}^2|\phi_3\rangle \\
 &= \frac{2^2}{\sqrt{15}}|\phi_1\rangle + \frac{11^2}{\sqrt{3}}|\phi_2\rangle + \frac{26^2}{\sqrt{5}}|\phi_3\rangle, \quad (\text{See Theorem A.1.}) \\
 &= \frac{4}{\sqrt{15}}|\phi_1\rangle + \frac{121}{\sqrt{3}}|\phi_2\rangle + \frac{676}{\sqrt{5}}|\phi_3\rangle.
 \end{aligned}$$

$$\begin{aligned}
\langle \psi | \hat{B}^2 | \psi \rangle &= \left(\frac{1}{\sqrt{15}} \langle \phi_1 | + \frac{1}{\sqrt{3}} \langle \phi_2 | + \frac{1}{\sqrt{5}} \langle \phi_3 | \right) \left(\frac{4}{\sqrt{15}} | \phi_1 \rangle + \frac{121}{\sqrt{3}} | \phi_2 \rangle + \frac{676}{\sqrt{5}} | \phi_3 \rangle \right) \\
&= \frac{4}{15} + \frac{121}{3} + \frac{676}{5} \\
&= \frac{4 + 605 + 2028}{15} \\
&= \frac{2637}{15} \\
&= \frac{879}{5}.
\end{aligned}$$

It follows that the expectation value of $\langle \hat{B}^2 \rangle$ is given by

$$\begin{aligned}
\langle \hat{B}^2 \rangle &= \frac{\langle \psi | \hat{B}^2 | \psi \rangle}{\langle \psi | \psi \rangle} \\
&= \frac{\left(\frac{879}{5} \right)}{3/5} \\
&= 293.
\end{aligned}$$

Exercises 2.7, 2.8, 2.9 and 2.10

All trivial.

For Exercise 2.7(b), you may wish to use Euler's identity, and for Exercise 2.8(e), the hyperbolic tangent identities may be useful. See, for instance, the formulae in [Tay05]

Exercise 2.11

Theorem. If $\hat{A}$ is projection operator, then $1 - \hat{A}$ is a projection operator.

Proof: Define the operator $\hat{B} = \hat{I} - \hat{A}$. As $\hat{A}$ is a projection operator by assumption, and by the properties of the adjoint, we have

$$\hat{B}^\dagger = (\hat{I} - \hat{A})^\dagger = \hat{I}^\dagger - \hat{A}^\dagger = \hat{I} - \hat{A} := \hat{B},$$

and

$$\hat{B}^2 = (\hat{I} - \hat{A})^2 = \hat{I} - 2\hat{A}\hat{I} + \hat{A}^2 = \hat{I} - 2\hat{A} + \hat{A} = \hat{I} - \hat{A} := \hat{B}$$

so that, by definition, $\hat{B}$ constitutes a projection operator, as required. ■

Appendix A

For the following result, I use the same notation as [Axl24].

Theorem A.1. Suppose $T \in \mathcal{L}(V)$, and v is an eigenvector of T corresponding to the eigenvalue λ , that is, $v \in V$ is such that $Tv = \lambda v$. Then for $n \in \mathbb{Z}^+$, we have $T^n v = \lambda^n v$.

Proof: The proof is by induction. Clearly the result is true for $n = 1$, so for our inductive step, suppose that this holds for some $n = k \geq 1$. Then, for $n = k + 1$, we have, by linearity,

$$T^n v = T(T^k v) = T(\lambda^k v) = \lambda^k T v = \lambda^k (\lambda v) = \lambda^{k+1} v = \lambda^n v,$$

which completes the proof. ■

References

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